RUTGERS–NEW BRUNSWICK
School of Arts and Sciences
Department of Physics and Astronomy



Eric Putney

Postdoctoral Associate · NHETC, Rutgers University

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APPOINTMENTS

Postdoctoral Associate · NHETC, Rutgers University September 2026 – Present

EDUCATION

Ph.D. in Physics · Rutgers, The State University of New Jersey August 2020 – May 2026
Thesis: Data Driven Methods for Measuring Galactic Dark Matter (defended May 2026)

B.S. *summa cum laude* Physics · The University of New Mexico September 2016 – May 2020

RESEARCH FOCUS

Physics-Informed Neural Networks for Cosmic Rays Ongoing
Paper in preparation *NHETC · Rutgers · New Brunswick, New Jersey*

- Built a physics-informed neural network (PINN) framework for Galactic cosmic-ray transport.
- Developed the training routine that enables fast forward modeling of cosmic-ray propagation and gamma-ray map calculations.

Dark QCD and the Hubble Tension September 2026
Cosmological Model Building · arXiv:2609.16120 *Rutgers · New Brunswick, New Jersey*

- Studied a dark-sector QCD model, in which confining dark radiation behaves as a viscous fluid, as a resolution of the Hubble tension.
- Ran CLASS, MontePython, Cobaya, and PROSPECT to evaluate the model against Planck, ACT, BBN, DESI DR2, Pantheon+, and SH0ES likelihoods.

Machine Learning for Galactic Dynamics January 2021 – May 2026
PhD Thesis · Advisor: Dr. Matthew Buckley · arXiv: 2205.01129, 2305.13358, 2412.14236, 2512.09989 *Rutgers*

- Fully data-driven measurement of the local Milky Way's gravitational potential and dark matter density.
- Training normalizing flows to learn the six-dimensional kinematic phase space of nearby stars from Gaia DR3 to solve the general equilibrium collisionless Boltzmann equation.
- Developed novel technique for unbiasing the kinematic phase space of stars obscured by dust extinction.
- Developed non-parametric Jeans analysis for benchmarking our flows technique in a hydrodynamically-simulated galaxy.

Neutrino Flavor Dynamics in Core-Collapse Supernovae October 2018 – May 2020
Undergraduate Honors Thesis · Advisor: Dr. Huaiyu Duan *UNM · Albuquerque, New Mexico*

- Analyzed the evolution of dense neutrino gases evolving in supernovae.
- Compared numerical simulations of neutrino oscillations in dense matter to analytic predictions.

PUBLICATIONS

September 2026 **Dark Radiation Sticks Together: Dark QCD and the Hubble Tension**
M. R. Buckley, N. Fernandez, **E. Putney** · hep-ph 2609.16120, INSPIRE

December 2025 **ClearPotential: Revealing Local Dark Matter in Three Dimensions**
E. Putney, D. Shih, S. H. Lim, M. R. Buckley · astro-ph 2512.09989, INSPIRE

December 2025 **ClearPotential: Learning the Dust-Corrected Potential of the Milky Way Using Flows**
E. Putney, D. Shih, M. R. Buckley, S. H. Lim · NeurIPS ML4PS 2025 Workshop, INSPIRE

December 2025 **GalaxyScore: Estimating Local Dark Matter Density in Galaxies with Score Matching**
S. H. Lim, D. Shih, M. R. Buckley, **E. Putney** · NeurIPS ML4PS 2025 Workshop, INSPIRE

December 2024 **Sweeping the Dust Away: Correcting the Phase Space Density of the Milky Way with Unsupervised Machine Learning**
E. Putney, D. Shih, S. H. Lim, M. R. Buckley · astro-ph 2412.14236

May 2023 **Mapping Dark Matter in the Milky Way using Normalizing Flows and Gaia DR3**
S. H. Lim, **E. Putney**, M. R. Buckley, D. Shih · JCAP, astro-ph 2305.13358

May 2022 **Measuring Galactic Dark Matter through Unsupervised Machine Learning**
M. R. Buckley, S. H. Lim, **E. Putney**, D. Shih · MNRAS, astro-ph 2205.01129

May 2020 **Numerical Analysis of Collective Neutrino Oscillations in Dense Neutrino Media**
Honors thesis (unpublished)

June 2019 **Measurement of the radial matrix elements for the $6s^2S_{1/2} \rightarrow 7p^2P_J$ transitions in cesium**
A. Damitz, G. Toh, **E. Putney**, C. E. Tanner, D. S. Elliott · Physical Review A, physics 1904.06362

PRESENTATIONS

October 2026 **Physics-Informed Neural Networks for Galactic Dark Matter**
Astronomy Seminar · Westlake University · Slides

October 2026 **Probing Dark Matter in the Milky Way with Physics-Informed Neural Networks**
Particle Seminar · Lawrence Berkeley National Laboratory · Abstract, Slides

September 2026 **Efficient PINN Models of Cosmic Rays**
2nd UNDARK Workshop, Instituto de Astrofísica de Canarias · Abstract

May 2026 **Data Driven Methods for Measuring Galactic Dark Matter**
Thesis Defense · Rutgers University, New Brunswick

May 2026 **Efficient PINN Models of Cosmic Rays**
Pheno 2026 Symposium, University of Pittsburgh · Abstract, Slides

December 2025 **ClearPotential: Learning the Dust-Corrected Potential of the Milky Way Using Flows**
Poster · NeurIPS ML4PS 2025 · Abstract, Letter, Poster

July 2025 **Gaia and Dark Matter**
Cargese Summer School · IESC · Indico

April 2025 **Model-free analyses of Gaia: mapping the local potential, dust, and dark matter**
YITP Seminar · Stony Brook University · Abstract

March 2025 **Model-free analyses of Gaia: mapping the local potential, dust, and dark matter**
Galaxy Hour · The Ohio State University · Abstract

March 2025 **Machine Learning and Galactic Dynamics**
Graduate Seminar · University of Cincinnati

November 2024 **Sweeping the Dust Away: An Unbiased Map of the Milky Way's Dark Matter and Gravitational Potential with Unsupervised Machine Learning**
ML4Jets 2024 · Abstract, Slides, Recording

May 2024 **Sweeping the Dust Away: An unbiased map of the Milky Way's gravitational potential using unsupervised ML**
DPF-PHENO 2024, University of Pittsburgh · Abstract, Slides

November 2023 **Mapping Dark Matter in the Milky Way using Normalizing Flows and Gaia DR3**
ML4Jets 2023, DESY · Abstract, Slides

May 2023 **Measuring Galactic Dark Matter through Unsupervised Machine Learning**
Phenomenology 2023 Symposium, University of Pittsburgh · Abstract, Slides

January 2023 **Measuring Galactic Dark Matter through Unsupervised Machine Learning**
241st AAS Meeting, Seattle WA · Abstract

April 2020 — **Numerical Analysis of Collective Neutrino Oscillations in Dense Neutrino Media**
Undergraduate Honors Thesis, University of New Mexico · Slides, Poster

August 2019 — **Laser Control for the Production of Excited Positronium in GBAR**
CERN Summer School · Slides

April 2019 — **Numerical Analysis of Collective Neutrino Oscillations in Core-Collapse Supernovae with Multidimensional Models**
University of New Mexico Physics Day · Slides

October 2018 — **Precision Measurement of the $6S \rightarrow 7P_{1/2}$ Cesium Transition Radial Matrix Element via Simultaneous Absorption Spectroscopy**
2018 APS Four Corners, University of Utah · Abstract, Slides

October 2017 — **Analysis of Diffusion of a Rhodium Adatom on a Tungsten (111) Surface**
2017 APS Four Corners, Colorado State University · Abstract, Poster

SCHOLARSHIPS AND AWARDS

2021 Rutgers Academic Fellowship
2020 Eion Gray Scholarship
2020 UNM Feynman Award
2019 Goldwater Scholar

SERVICE

Rutgers Chapter of the AAUP-AFT
Graduate Worker Organizer

- Physics and Astronomy Department Representative — December 2022 – 2026
- Fellows Organizing Committee Chair — January 2024 – December 2025
- Graduate Steering Committee and Executive Council Member — September 2024 – 2026
- Short Term Staff Organizer for Clinically Focused University Practitioners — January 2025 – June 2025

Rutgers Physics & Astronomy Graduate Student Organization
President — September 2023 – June 2024

Rutgers Graduate Student Association
Treasurer — January 2025 – June 2025

REFERENCES

Dr. Matthew Buckley
Rutgers, The State University of New Jersey, Department of Physics and Astronomy

Dr. David Shih
Perimeter Institute for Theoretical Physics and University of Waterloo

Dr. Tracy Slatyer
Massachusetts Institute of Technology, Center for Theoretical Physics

Dr. Sung Hak Lim
Institute for Basic Science, CTPU-PTC